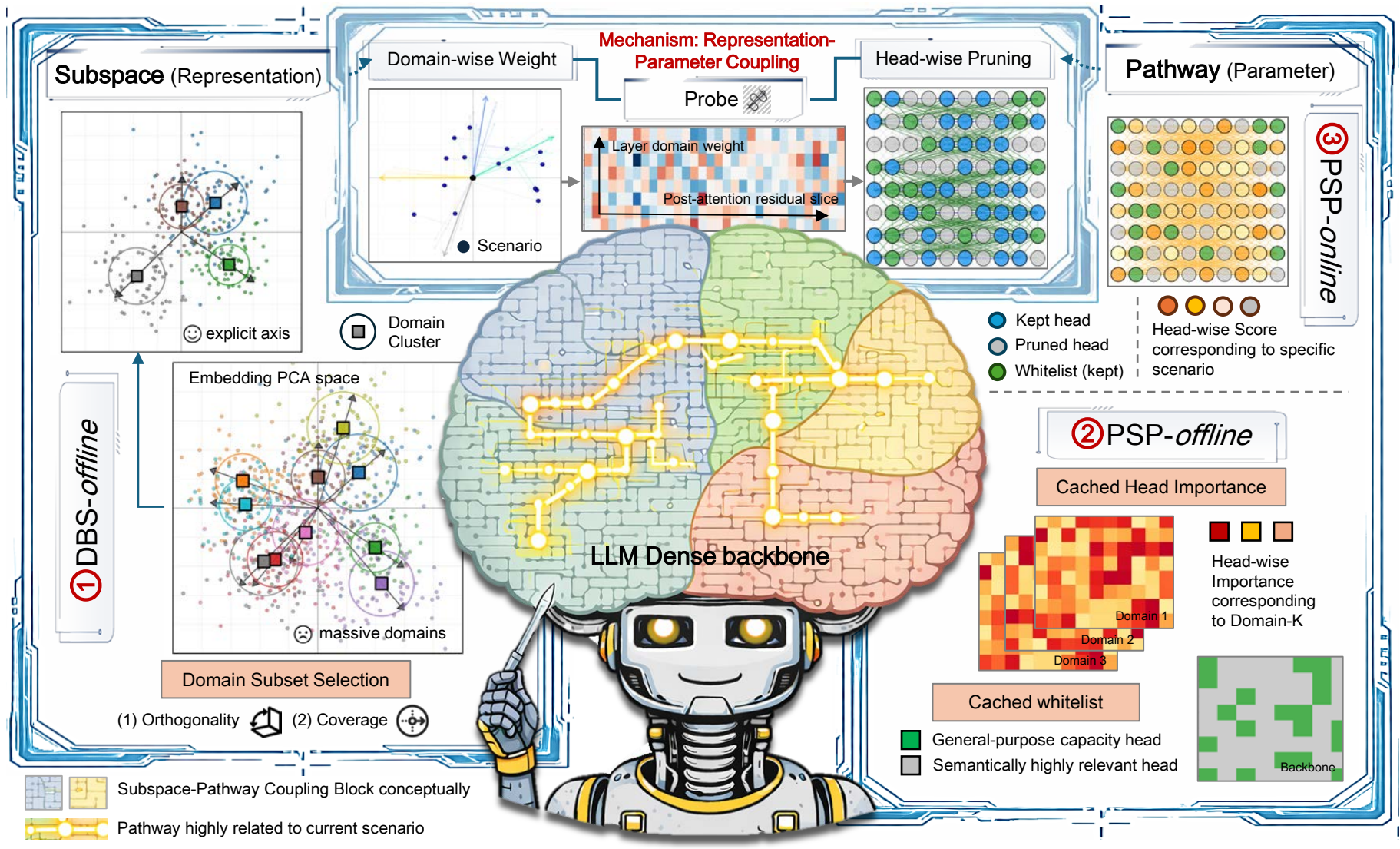
Subspace (Representation)
Mechanism: Representation-Parameter Coupling
Domain-wise Weight
Probe
Head-wise Pruning
Pathway (Parameter)
Scenario
Layer domain weight
Post-attention residual slice
③ PSP-online
explicit axis
Domain Cluster
Kept head
Pruned head
Whitelist (kept)
Head-wise Score corresponding to specific scenario
Embedding PCA space
① DBS-offline
massive domains
② PSP-offline
Cached Head Importance
LLM Dense backbone
Domain 1
Domain 2
Domain 3
Head-wise Importance corresponding to Domain-K
Domain Subset Selection
(1) Orthogonality
(2) Coverage
Cached whitelist
General-purpose capacity head
Semantically highly relevant head
Backbone
Subspace-Pathway Coupling Block conceptually
Pathway highly related to current scenario